

Abu Moaz

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Professional Summary

[HTTPS://ABUMOAZ.ME](https://abumoaz.me)

Python Backend & AI Engineer with 1 year of experience building production RAG systems, scalable FastAPI services, and search infrastructure. Led zero-downtime migration of 26M+ record Elasticsearch cluster and shipped AI ticketing assistants using AWS Bedrock and pgvector. Currently building Validgen (validgen.com), an AI code security platform with 14+ active users.

Technical Skills

Languages: Python, JavaScript, Java

Frameworks: FastAPI, Flask, React.js, Node.js

AI & LLM: RAG, LangChain, LlamaIndex, Hugging Face Transformers, AWS Bedrock, OpenAI API

Databases & Search: PostgreSQL, pgvector, Elasticsearch, Redis

Infrastructure: AWS (EC2, S3, IAM), Docker, GitHub Actions, CI/CD, Linux

Tools: Git, REST API Design, JWT/OAuth, Boto3, Microservices

Experience

Blackcoffer Pvt Ltd

June 2025 – Feb 2026

Software Engineer

Noida, UP

- Designed and developed scalable backend services using FastAPI to process high volume REST API requests with optimized latency.
- Led zero downtime migration of search infrastructure handling 26M+ records from Elasticsearch 5.6 to 9.2, implementing backward compatible APIs and minimizing client side migration effort by 95%.
- Built a legal document automation platform with role-based modules, enabling admins to create and manage document templates while allowing users to securely purchase, customize, and generate legal documents.
- Developed a multi role AI ticketing assistant, implementing RAG pipelines using pgvector and cosine similarity utilizing strict evals to ensure retrieval accuracy and context aware historical resolutions with verifiable citations based on specific user permissions.
- Provisioned AWS cloud deployments using EC2, S3, and IAM, utilizing the boto3 SDK to orchestrate AWS Bedrock foundational models within core pipelines.

Frazor Enterprises Solutions

Feb 2025 – May 2025

Associate Software Engineer (Internship)

Gurgaon, Haryana

- Architected high throughput REST APIs using FastAPI, establishing integration with frontend environments to support core application functionality.
- Managed third party API integrations with continuous data synchronization, implementing distributed rate limiting using a sliding window algorithm to ensure system stability under high request loads.
- Developed asynchronous inference endpoints using FastAPI to integrate generative AI models, optimizing payload processing and backend response times.
- Contributed to data ingestion pipelines to process and format raw user inputs, implementing chunking strategies to ensure context window management and controlled token usage for cost efficiency and reduced hallucinations.

Validgen (Indie Project)

Feb 2026 – Present

Solo Founder / AI Backend Engineer

Remote

- Built and deployed an AI code security scanner (validgen.com) utilizing 150+ custom detection rules combining regex and AST analysis to identify vulnerabilities in AI-generated code.
- Architected secure scanning environment using Docker-in-Docker with ephemeral containers, enforcing zero code retention policy post-analysis.
- Acquired 14 users and processed 50+ repository scans within the first month by targeting security flaws in code generated by AI agents.
- Designed AI-driven remediation pipeline mapping detected vulnerabilities to predefined rules for automated code resolution.

Projects

- Validex – API Contract Validator** | *FastAPI, PostgreSQL, pgvector, Docker, Redis, Open-Source LLM*

github

 - Developed an automated testing platform utilizing FastAPI to ingest OpenAPI specifications, implementing RAG pipelines with pgvector to generate accurate integration test scripts via open-source large language models.
 - Architected a secure execution environment utilizing Docker to run generated test scripts within isolated sandboxed containers, ensuring safe interaction with target APIs.
 - Orchestrated asynchronous task processing utilizing Redis to manage continuous test generation and container execution queues, optimizing overall system throughput for complex validations.
 - Constructed a comprehensive contract validation system that accelerates the software development lifecycle by automating the creation and execution of extensive API test coverage.
- Prepify – AI-Driven Entrance Prep** | *Next.js, PostgreSQL, FastAPI, LlamaIndex, pgvector, Ollama, Docker*

prepify

 - Built a generative AI assessment pipeline using LlamaIndex to ingest study material and generate context-aware MCQs guided by patterns from previous year competitive exam questions.
 - Architected semantic retrieval layer using pgvector within PostgreSQL to store and query high-dimensional embeddings for accurate context selection.
 - Implemented real-time quiz workflows and personalized practice modules with role-based access control for dynamic, adaptive learning experiences.

Education

Aligarh Muslim University
Master Of Computer Applications (CGPA: 8.5 / 10)

June 2025
Aligarh, UP